Study Design and Analysis Workflow

Primary EGFR Prediction Cohort

NSCLC Radiogenomics cohort
Public CT imaging, tumor segmentation,
clinical data, and EGFR labels

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Final analyzable cohort
N = 97 patients
23 EGFR-mutant / 74 EGFR-wildtype

↓

CT–SEG matching and tumor-mask reconstruction
DICOM spatial correspondence QC

↓

Radiomics extraction
107 Original PyRadiomics features
1-mm isotropic resampling; bin width = 25 HU

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Repeated nested cross-validation
20 repetitions × 5 outer folds
4-fold inner tuning
Unweighted L2 logistic regression

↓

Primary and comparator models
Radiomics-only | Clinical-only | Protocol-only
Radiomics + Clinical
ROC-AUC, PR-AUC, and Brier score

↓

Feature stability across 100 fitted outer models

↓

18 stable radiomics features

Independent Test–Retest Cohort

RIDER Lung CT
Independent test–retest dataset

↓

28 patients
56 CT scans
TEST + RETEST

↓

Manual primary-tumor
SEG reconstruction

↓

Same radiomics pipeline
107 features

↓

Independent reproducibility analysis
ICC(2,1), absolute agreement

↓

Reproducibility results
14/18 features: ICC ≥ 0.75
8/18 features: ICC ≥ 0.90
Median ICC = 0.870